MIL-S-45669A(MU)
19 February 1971
SUPERSEDING
MIL-S-45669(ORD)
5 May 1960

MILITARY SPECIFICATION

SIGHT, BORE, OPTICAL: M45

1. SCOPE

1.1 This specification covers one type of instrument known as Sight, Bore, Optical: M45 used for boresighting mortars. Throughout this specification, this instrument is referenced as “Bore Sight”. elbow Telescope, 8579641-1 is an integral component of this bore sight.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-13321	Telescope, Elbow, M62 (Series)
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

FSC 1240

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DRAWINGS

U.S. Army, Frankford Arsenal

F8579189	Sight, Bore, Optical: M45

PACKAGING DATA SHEET

5800859	Packaging of Sight, Bore, Optical: M45 W/E

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- The materials shall be in accordance with the drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The bore sight shall be manufactured in accordance with Drawing F8579189.

3.3 General specifications.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926.

3.3.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.4 Telescope, Elbow, 8579641-1.- The Elbow Telescope, 8579641-1, which is a part of the bore sight, shall have met the requirements of MIL-T-13321 prior to assembly as a bore sight.

3.5 Vibration.- The bore sight shall meet all requirements of this specification after being positioned as in (a) of Figure 1 and vibrated at a constant frequency of 30 cycles per second (cps) with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds.

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3.6 Orientation.- The bore sight shall be clamped securely in each of the three mounting positions shown in Figure 1 with the respective V slide contacting the longitudinal portion of a smooth 4.5 inch diameter tube or equivalent, whose axis is inclined precisely 45 degrees above the horizonal. The following conditions shall be met at each of the above specified positions unless otherwise specified.

3.7 Accuracy.

3.7.1 Collimation.- The line of sight of the elbow telescope shall be parallel to the vertical plane passing through the axis of the corr-esponding V slide within 3 roils for positions (a) and (c) of Figure 1.

3.7.2 Plumb travel.- Using position (a) of Figure 1 in accordance with 3.6 and with the line of sight of the elbow telescope in a hori-zontal plane and in coincidence with a plumb line, the line of sight shall not deviate from the plumb line by more than 2.0 roils total dis-placement from maximum depression to maximum elevation.

3.7.3 Parallelism of reticle and image.- The vertical crossline on the reticle shall be parallel to the image of a plumb line within 1 degree.

3.7.4 Cross level vial.- The respective cross level bubble shall be central within 5 minutes.

3.7.5 Elevation level vial.- The elevation level bubble shall be central within 5 minutes.

3.8 Telescope travel.- With position (a) of Figure 1 established in accordance with 3.6, the telescope shall be capable of being rotated a minimum of 45 degrees in elevation and 30 degrees in depression.

3.9 Operating temperature.- The bore sight shall meet the torque requirements of 3.10 when thermally stabilized at ambient temperatures of minus 40 degrees Fahrenheit and plus 150 degrees Fahrenheit and the strap assembly shall be completely operable when securing and loosen-ing the bore sight to the tube.

3.9.1 Clamp screw.- The clamp screw, in a free state, shall be op erable throughout its threaded portion without restriction or binding at all temperatures specified above.

3.10 Torque.- The torque required to elevate and depress the tele-scope through full travel shall not exceed 10 pound inches nor be less than 2 pound inches at room temperature (60 to 90 degrees). The torque required to elevate and depress the telescope through full travel shall not exceed 28 pound inches nor be less than 2 pound inches at the extreme temperatures specified in 3.9.

3.11 Strap assembly.- The bore sight or strap assembly shall not become loosened as a result of applying 2 pounds direct force at the knurled portion of the clamping screw once in each opposing direction

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parallel to the tube axis, when the bore sight is positioned and secured as shown in (a) of Figure 1 with a maximum torque of 2 pound inches applied to the clamp screw;

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government-reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirement of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 bore sights plus 3 each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of Product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions

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of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Table I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR : AQL 0.65% Defective			
101.	Vibration	3.5	4.6.1
102.	Orientation	3.6	4.6.2
103.	Collimation	3.7.1	4.6.3
104.	Parallelism of reticle and image	3.7.3	4.6.5
105.	Cross level vial	3.7.4	4.6.6
106.	Elevation level vial	3.7.5	4.6.7
107.	Telescope travel	3.8	4.6.8
108.	Torque	3.10	4.6.11
109.	Strap assembly	3.11	4.6.12
110.	Plumb travel	3.7.2	4.6.4
MINOR:	None Defined		

NOTE: The tests in Table I shall be conducted at a temperature between +60 and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with “Acceptance and Rejection” as specified in MIL-STD-105.

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4.4 Special sampling.

4.4.1 General.- Three (3) bore sights shall selected at random as a special sample from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURES
301.	Materials	3.1	Applicable Dwgs - Visual
302.	Fabrication	3.2	Applicable Dwgs - Visual
303.	General specification	3.3	MIL-F-13926 - Visual

4.4.2 Environmental.- Three (3) bore sights shall be selected at random as a special sample from each 50 produced or from each month's production, whichever occurs first. The samples shall have met the requirements and tests in Table III and shall then meet the requirements and tests in Table I.

TABLE III

	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURES
304.	Operating temperature	3.9	4.6.9
305.	Clamp screw	3.9.1	4.6.10
306.	Torque	3.10	4.6.11

NOTE: The tests in Table III shall be conducted at temperatures of minus 40 degrees Fahrenheit (-40°F) and plus 150 degrees Fahrenheit (+150°F).

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4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Vibration.- The bore sight shall be vibrated by means of a device capable of meeting the requirements of 3.5. With the aid of a suitable adapter, the Bore Sight, M45 shall be secured firmly to the device and vibrated in the position (a) indicated on Figure 1.

4.6.2 Orientation.- orientation is accomplished by means of a special testing device to assure correct orientation in the three positions indicated on Figure 1. The device shall consist of a simulated tube, with a base capable of supporting the tube, at an angle of 45 degrees plus or minus 20 seconds. The outside diameter of the tube shall

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measure 4.500 inches plus .050 inch and parallelism shall be within .003 inch per foot. The surface quality shall be between 32 and 63 micro inches RMS.

4.6.3 Collimation.- This test is accomplished by means of two plumb lines located approximately fifty feet apart, and the tube of the device positioned midway between the plumb lines so that the axis of the tube lies in the plane of the plumb lines. This can be checked by locating a vertical marker on the center line of the tube at each end. By means of a collimator, located approximately 15 feet from the back of the plumb line, bring the two plumb lines and the two markers on the tube in alinement with the vertical line of the collimator. A peep sight placed on the objective end of the collimator, with a 3/16 inch aperture will enable the observer to make an accurate reading. Mount the bore sight on the tube ,as illustrated in (a) of Figure 1, by means of the strap assembly. Rotate the bore sight around the outside diameter of the tube until the cross level bubble is centered, and tighten the clamp screw. View through the telescope the image of the plumb line. The vertical line of the telescope shall coincide with the plumb line within the tolerance indicated in 3.7.1 as read on a mil chart placed in back of the plumb line. The bore sight is then reversed on the tube, as illustrated in (c) of Figure 1, and the same cross level and viewing procedure employed, using the opposite plumb line.

4.6.4 Plumb travel.- Plumb travel check consists of positioning the bore sight and plumb line as described in 4.6.6. The plumb line as viewed through the telescope at the intersection of the cross lines of the reticle shall not deviate more than the tolerance of 3.7.2 when the telescope is displaced from maximum elevation to maximum depression. A mil block chart, located at one end of the plumb line, will indicate the amount of deviation.

4.6.5 Parallelism of reticle and image.- Parallelism of reticle and image check is made with the bore sight and plumb line positioned as described in 4.6.6. Locate a mark on a chart in back of the plumb line which will indicate the horizontal line of sight of the telescope. Move the telescope to a position where the horizontal line of the reticle will coincide with the mark. Observe the vertical line of the telescope in relation to the plumb line which shall be within the tolerance of 3.7.3. A degree chart located in back of the plumb line will indicate the mount of deviation.

4.6.6 Cross level vial.- Cross level vial check for accuracy consists of mounting the bore sight on the tube of the device by means of the strap assembly in the position (a) of Figure 1. Locate a plumb-line precisely eight feet from the center of the pivot shaft of the bore sight and in a vertical plane with the axis of the tube. By means of a peep sight with a 1/8 inch aperture placed on the objective end of the telescope, move the bore sight until the cross mark of the telescope coincides with the plumb line the full excursion of elevation and depression. In this position, check the cross level vial which shall meet the requirements of 3.7.4.

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4.6.7 Elevation level vial.- Elevation level vial test is made by checking the 45 degree angle of the tube of the device by means of a commercial optical protractor. Mount the bore sight on the tube as illustrated in (a) of Figure 1, and cross level. In this position, the elevation level vial shall be central within the tolerance of 3.7.5. The same test is repeated for checking the other "V" slide with the bore sight mounted as illustrated in (c) of Figure 1.

4.6.8 Telescope travel.- Telescope travel is accomplished with the bore sight and plumb line located as indicated in 4.6.6. Locate a mark on a chart eight feet above the horizontal line of sight which will indicate 45 degree elevation point, and another mark feet 7-1/2 inches below the horizontal line of sight which will indicate the 30 degree depression point. The telescope shall travel from one point to the other to meet requirements of 3.8.

4.6.9 Operating temperature.- Operating temperature test is made by clamping the bore sight to a 4.50 inch outside diameter steel tube by means of the strap assembly and exposing it to temperatures indicated in 3.9 by means of a commercial cold chest and oven. When thermally stabilized at these temperatures, the torque requirement of 3.10 shall be made by means of a commercial spring scale attached to the end of the telescope housing. The operability of the strap assembly is then checked by releasing the clamp and moving the bore sight, and re-clamping at different positions.

4.6.10 Clamp screw.- Clamp screw check is a manual "test made by revolving the screw from one end of the thread to the other end without binding or other restrictions at all temperatures herein specified.

4.6.11 Torque.- Torque test shall be made by means of a spring scale which can be hooked at one end of the telescope housing. The telescope is then moved through its full travel by pulling the scale and observing the indicator of the scale for requirements of 3.10.

4.6.12 Strap assembly.- Strap assembly test is performed by mounting the bore sight on the tube of the device in (a) of Figure 1, and by means of a torque wrench and adapter, apply a pressure of two pound inches to the clamp screw. Sight the telescope on a target at the intersection of the crosslines of the telescope. Hook a spring scale on the knurled portion of the clamp screw and apply the pressure indicated in 3.11 in opposing directions parallel to the tube axis. Observe the target thru the telescope and there shall be no relative movement of the telescope.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 5800859. The level of protection shall be as specified in the procurement document-

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6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable stock number.
(d) Provisions for First Article Testing.

Custodian:

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Preparing activity:

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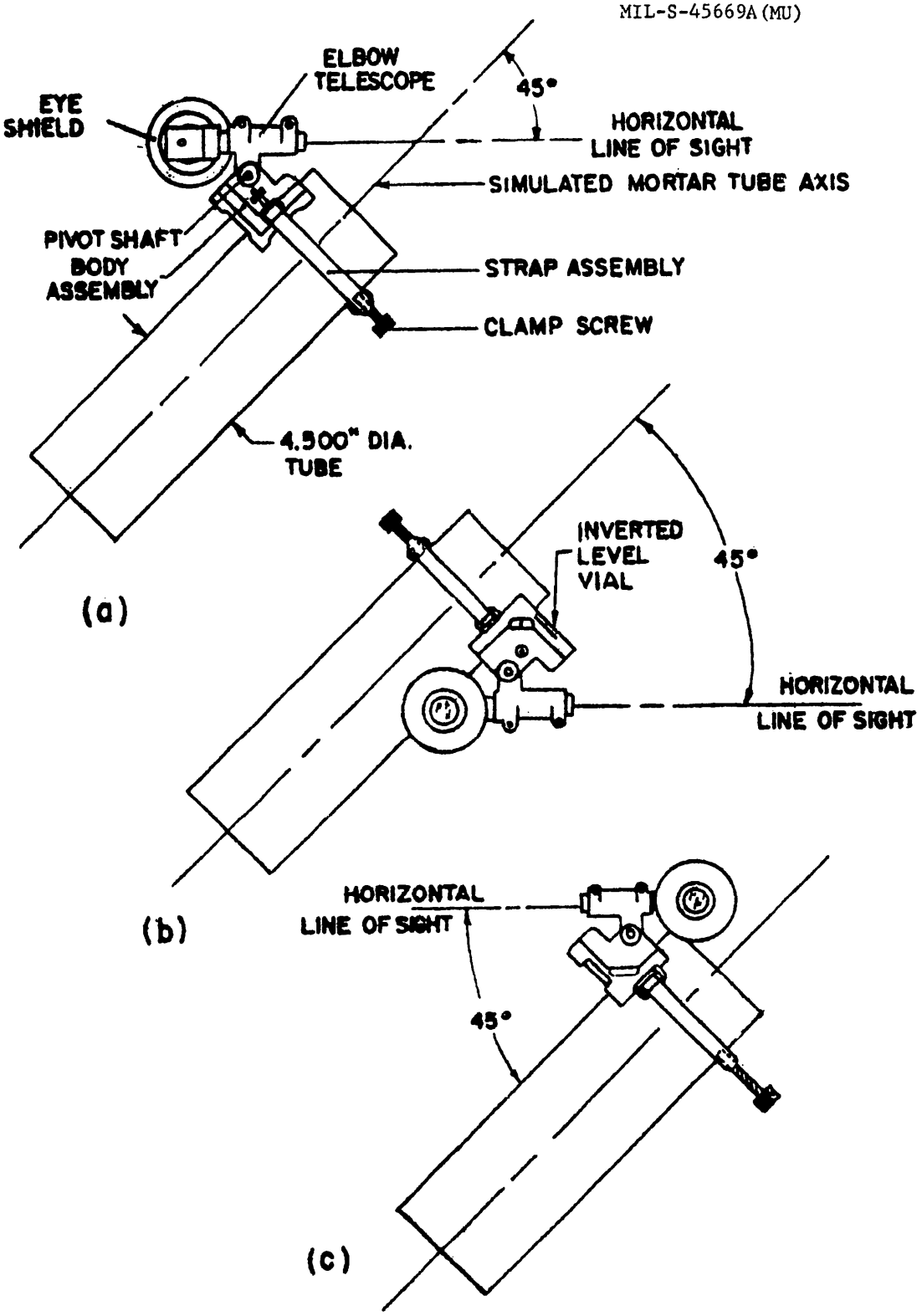


FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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